

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 02

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Recap of Previous Lecture



Topic

Instruction

Topic

ISA

Topic

Types of Instruction

Topics to be Covered



Topic

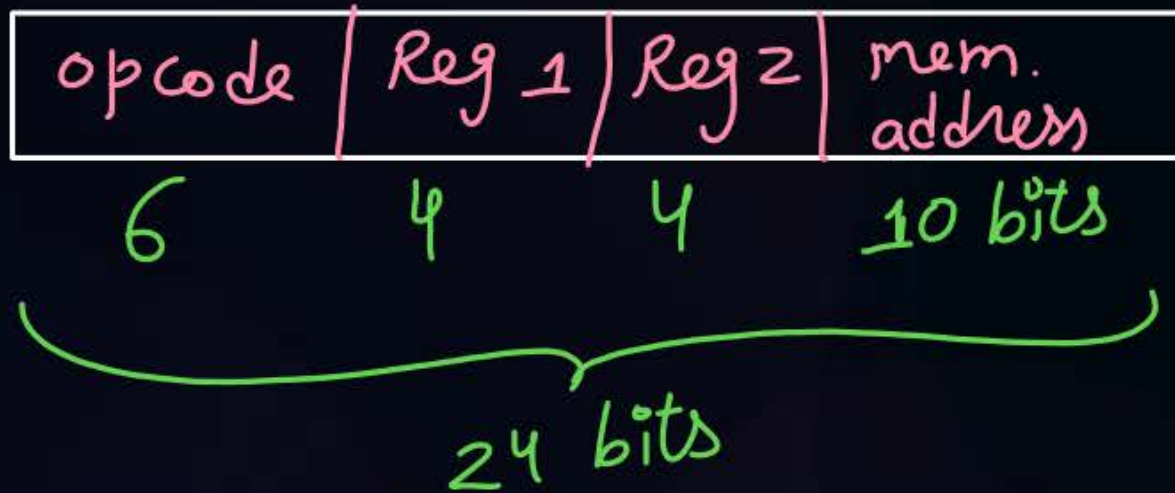
Instruction

Topic

Multiple Instruction Support

$$\text{opcode} = \lceil \log_2 50 \rceil = 6 \text{ bits}$$

#Q. A processor has 50 distinct instructions and 16 general purpose registers. Each instruction in system has one opcode field, 2 register operand field and a 10 bits memory address field. The length of the instruction is ____ bits? Ans = 24



no. of registers = 16

Reg. number = 4 bits

$$= \lceil \log_2 16 \rceil = 4 \text{ bits}$$

$R0 \Rightarrow 0000$
 $R1 \Rightarrow 0001$
 $\vdots$
 $R15 \Rightarrow 1111$

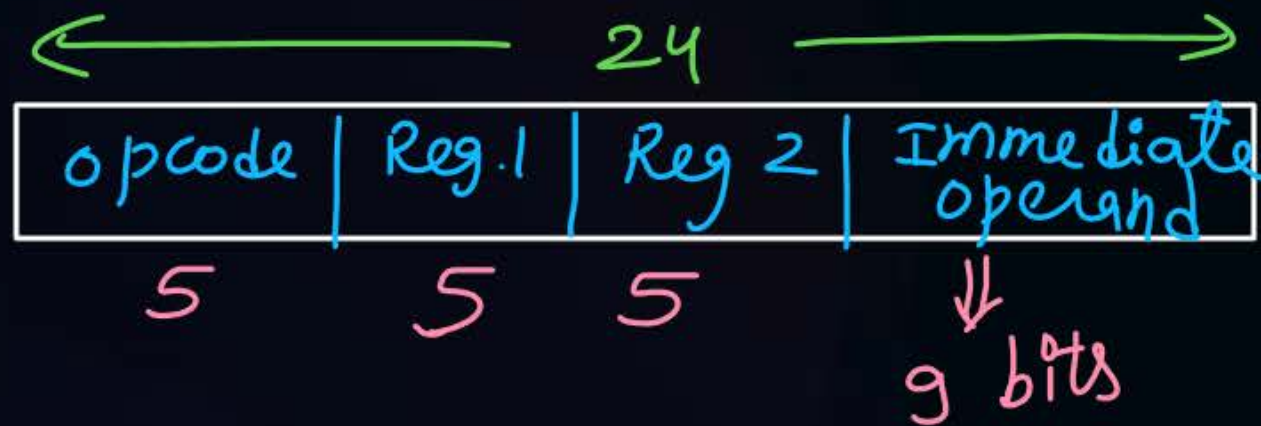
[NAT]



$$\text{opcode} = \lceil \log_2 20 \rceil = 5 \text{ bits}$$

$$\text{Reg} = 5 \text{ bits}$$

#Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9?



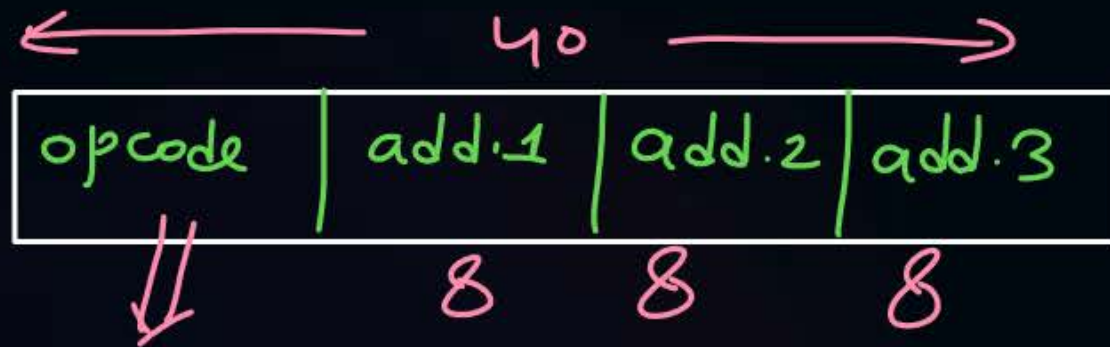
#Q. A processor has 20 distinct instructions and 32 general purpose registers. A 24-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 9?

In above question: Assume that immediate operand field is an unsigned number, What is its maximum and minimum value possible?

$$\begin{aligned} \text{min} &= 0 \\ \text{max} &= 2^9 - 1 = 511 \end{aligned}$$

for n bits		min	max
Unsigned		0	$2^n - 1$
Signed	sign-magnitude	$\rightarrow -\left(2^{n-1} - 1\right)$	$+ \left(2^{n-1} - 1\right)$
	1 st complement		
	2 nd complement	-2^{n-1}	$+ \left(2^{n-1} - 1\right)$

#Q. Consider a system which support only 3 address instructions only, and supports 256B memory. If the instruction size is 40-bits then maximum & minimum number of instruction supported by the system are?



$$40 - 24 = 16 \text{ bits}$$

$$\begin{aligned} \text{max} &= 2^{16} = 65536 \\ \text{min} &= 1 \end{aligned}$$

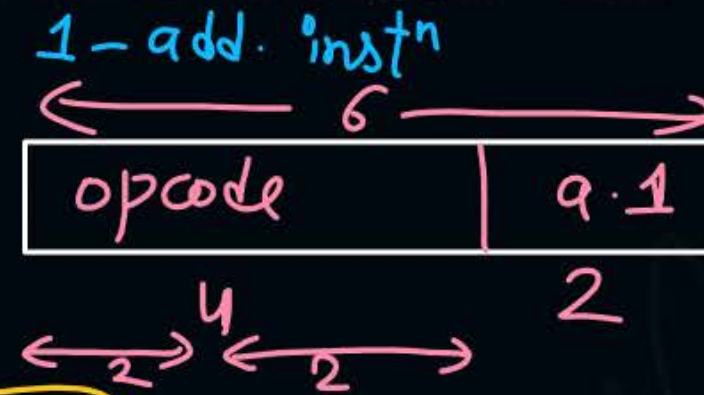
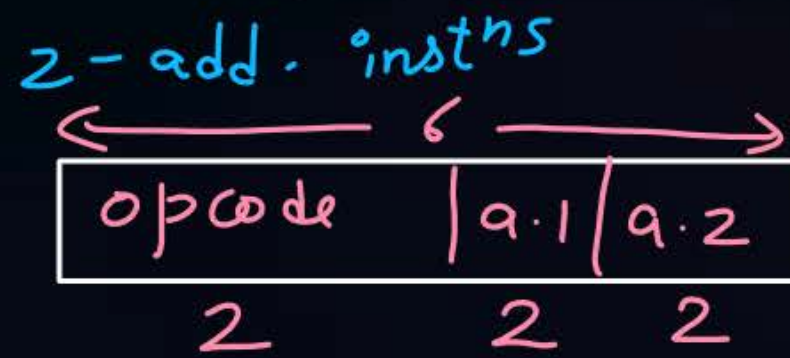
$$\text{mem.} = 256 \text{ B}$$

$$\text{no. of cells} = \frac{256 \text{ B}}{1 \text{ B}} = 256 = 2^8$$

$$\text{address} = 8 \text{ bits}$$

Ans = 4

#Q. Consider a computer which supports only 2-address and 1-address instructions. Each instruction is of 6-bits and each address is of 2-bits. If there are 3 2-address instructions supported by the system then maximum number of 1-address instructions supported by system is?

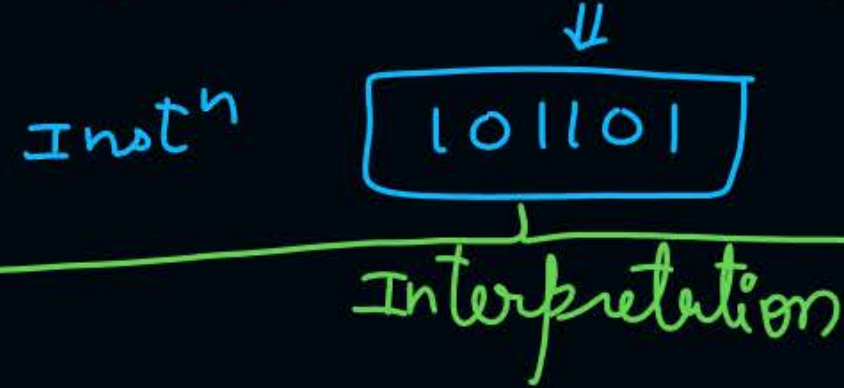


max opcode combinations = $2^2 = 4$ (00, 01, 10, 11)
 used — 11 — = 3 (assume: 00, 01, 10)
 unused " " = 1 (as per above assumption ⇒ 11)

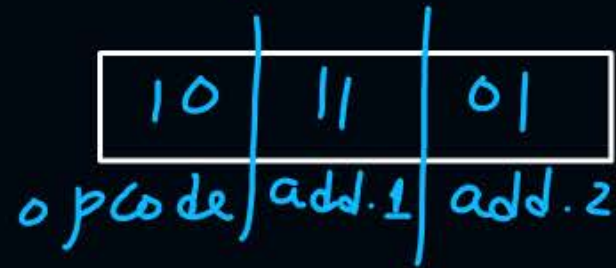
1100
 1101
 1110
 1111

max 1 add. instⁿs = 4

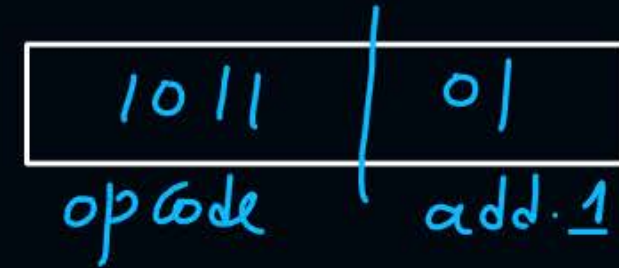
Assume, 6 bits instⁿ created by compiler.
It comes to CPU for execution



✓ as 2-add. instⁿ



✗ as 1-add. instⁿ



Inst ^{ns}	treated as
001100	2-add.
110000	1-add.

2-add. inst ^{ns}	unused opcodes	max 1 add. inst ^{ns}
4	$4 - 4 = 0$	$0 * 2^2 = 0$
3	$4 - 3 = 1$	$1 * 2^2 = 4$
2	$4 - 2 = 2$	$2 * 2^2 = 8$
1	$4 - 1 = 3$	$3 * 2^2 = 12$
0	$4 - 0 = 4$	$4 * 2^2 = 16$

only 2-add. inst^{ns}
supported

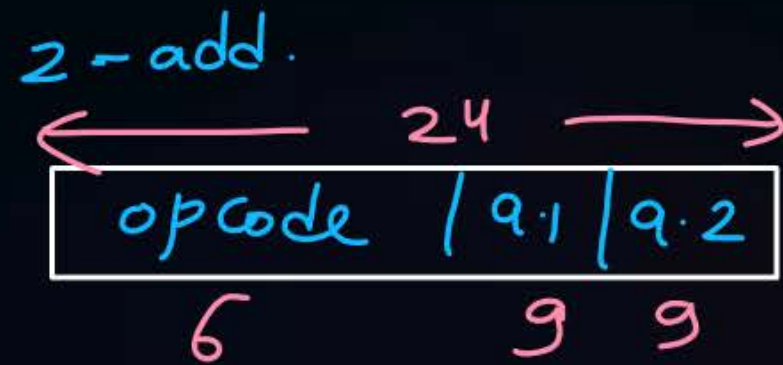
only 1-add. type
inst^{ns} supported.

#Q. Consider a computer which supports only 2-address and 1-address instructions. Each instruction is of 6-bits and each address is of 2-bits. If there are 3 2-address instructions supported by the system then maximum number of 1-address instructions supported by system is?

In above instruction what is the range of number of 1-address instructions supported?

1 to 4

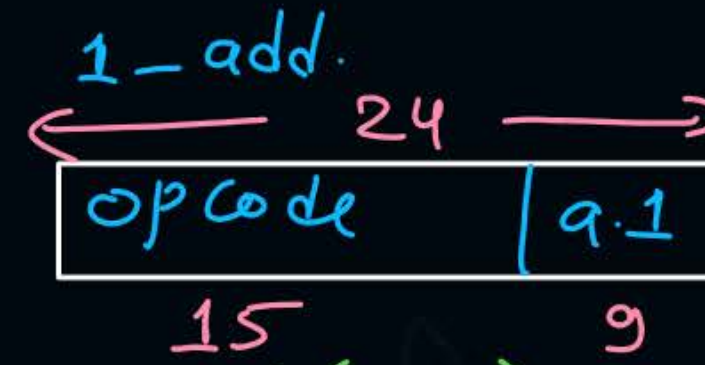
#Q. Consider a system with 24-bit instructions and 9-bit addresses. If there are 57 2-address instructions then maximum how many 1-address instructions can be formulated in the system?



$$\text{max opcodes} = 2^6 = 64$$

$$\text{used opcodes} = 57$$

$$\text{unused opcode} = 7$$

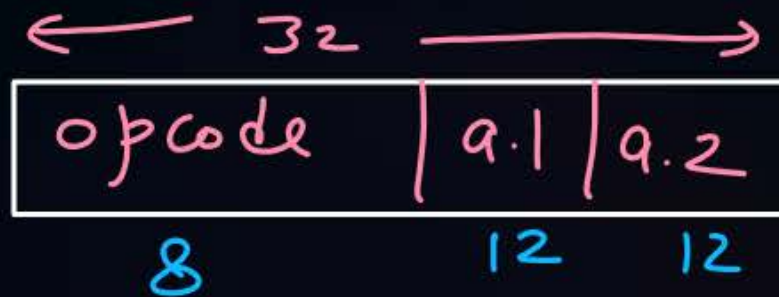


$$\text{max} = 7 * 2^9 = \underline{\underline{3584}} \text{ Ans.}$$

Ans = 8192

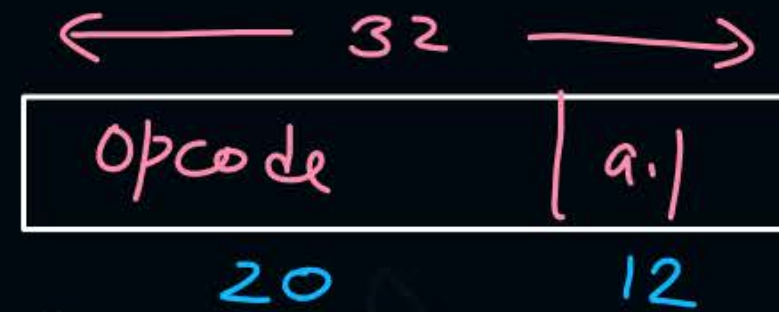
#Q. Consider a system with 32-bit instructions and 12-bit addresses. If there are 254 2-address instructions then maximum how many 1-address instructions can be formulated in the system?

2-add.



$$\begin{array}{rcl}
 \text{max} & = & 2^8 = 256 \\
 \text{used} & = & 254 \\
 \hline
 \text{unused} & = & 2
 \end{array}$$

1-add.

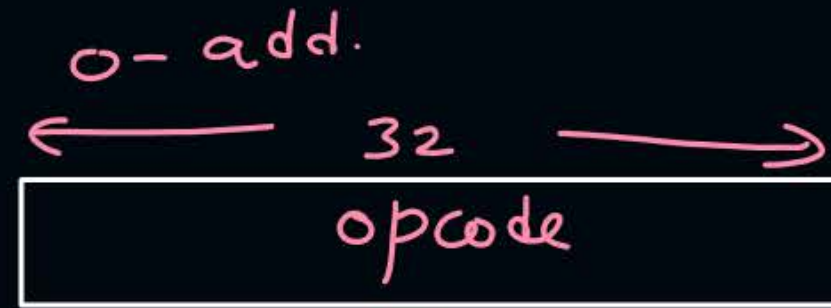


$$2 * 2^{12} = 8192$$

$$\text{Ans} = 28672$$

#Q. Consider a system with 32-bit instructions and 12-bit addresses. If there are 254 2-address instructions and 8185 1-address instructions then maximum how many 0-address instructions can be formulated?

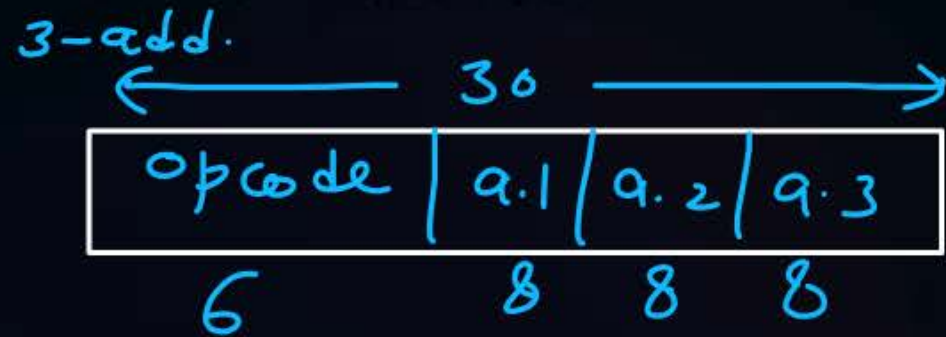
$$\begin{array}{r} \text{max 1 add. inst}^{\text{ns}} = 8192 \\ \text{used} = 8185 \\ \hline \text{unused} = 7 \end{array}$$



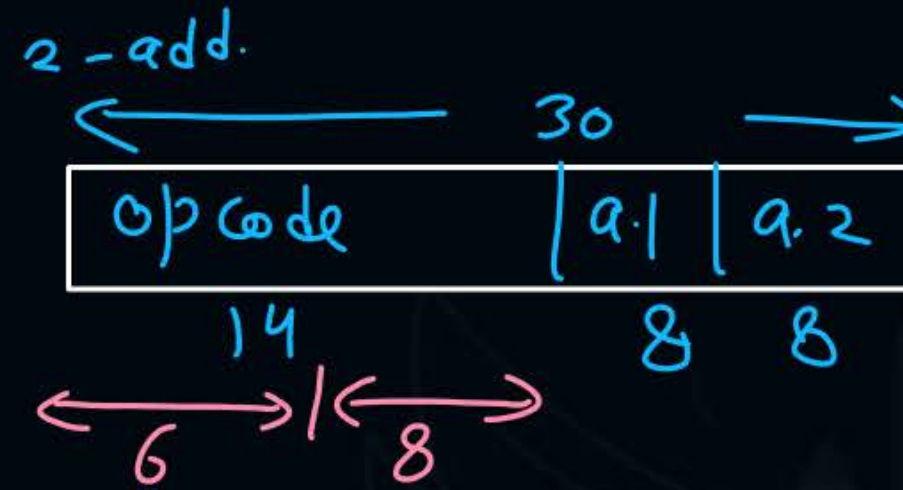
$$\text{max} = 7 * 2^{12} = 28672$$

$$\text{Ans} = (64 - x) 2^8$$

#Q. Consider a system which supports 3-address and 2-address instructions both. It has 30-bit instructions with 8-bit addresses. If there are 'x' 3-address instructions then maximum how many 2-address instructions can be formulated?



$$\begin{aligned} \text{max} &= 2^6 = 64 \\ \text{used} &= x \\ \hline \text{unused} &= 64 - x \end{aligned}$$



$$(64 - x) * 2^8$$

#Q. In previous question if there are maximum 1024 2-address instructions then the value of x is 60?

$$(64 - x) 2^8 = 1024$$

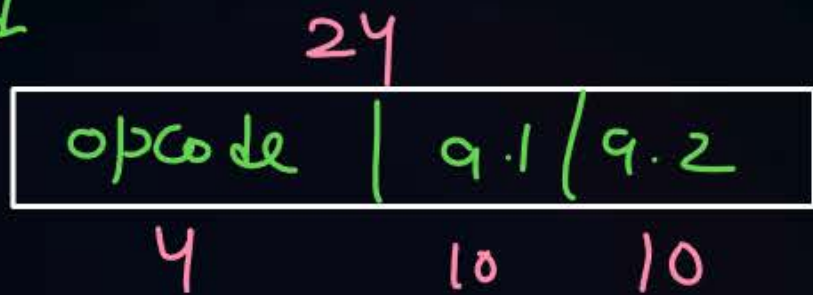
$$64 - x = 4$$

$$\boxed{x = 60}$$

Ans = 12

#Q. Consider a system which supports 2-address and 1-address instructions both. It has 24-bit instructions with 10-bit addresses. If there are 4096 1-address instructions then maximum how many 2-address instructions can be formulated?

2-add.

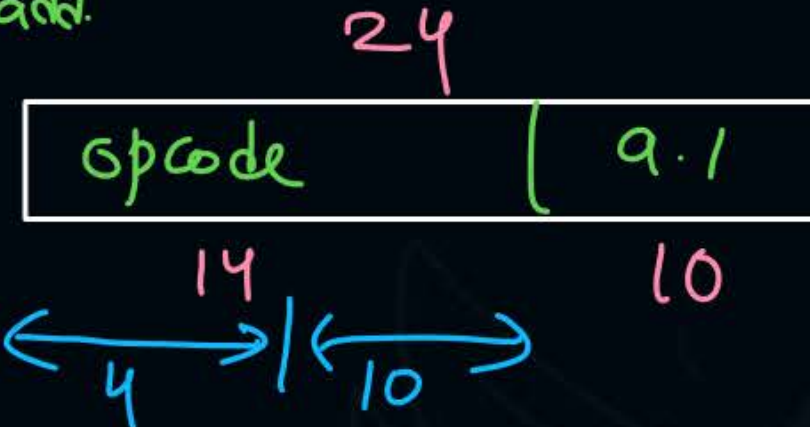


$$\text{max} = 2^4 = 16$$

$$\text{used} = x$$

$$\text{unused} = (16 - x)$$

1-add.



$$(16 - x) * 2^{10} = 4096$$

$$16 - x = 4$$

$$\boxed{x = 12}$$

Ans = 48

Reg = 6 bits

#Q. Consider a system with 16-bits instructions and 64 CPU registers. The System supported 2 types of instructions: Type-A and Type-B.

Type-A instructions have an opcode, one register operand and one immediate operand of 3-bits

Type-B instructions have an opcode, and 2 register operands.

If there are 10 Type-B instructions supported by the system then maximum how many Type-A Instructions supported by the system?

Type - A



Diagram showing the calculation of the maximum number of instructions for Type A:

4 (bits for opcode) and 3 (bits for register) are indicated by arrows pointing to the calculation:

$$\text{max} = 6 * 2^3 = \underline{\underline{48}} \quad \text{Ans.}$$

Type - B



Diagram showing the calculation of the maximum number of instructions for Type B:

$$\begin{aligned} \text{max} &= 2^4 = 16 \\ \text{used} &= 10 \\ \hline \text{unused} &= 6 \end{aligned}$$

- #Q. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____?

H.W.

#Q. A processor has 16 integer registers ($R_0, R_1, \dots, R_{15}$) and 64 floating point registers ($F_0, F_1, \dots, F_{63}$). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type 4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating-point register operand (1F).

The maximum value of N is _____ ?



2 mins Summary



Topic

Instruction

Topic

Multiple Instruction Support





Happy Learning

THANK - YOU

